
Unit – 2 : Session – 8 : SLO – 2 | Issues in Multithreaded Programming — Scenario Diagnosis

Scenario 1: Shared Counter

Q1. Threading Issue: Race condition.

Q2. Reason: counter++ is not atomic (load, add, store), so the two threads interleave and overwrite each other's updates; the result depends on thread timing, giving 19,500 or 20,300 instead of 20,000.

Q3. Solution: put the increment in a critical section protected by a mutex (or semaphore), or use an atomic increment.

Scenario 2: Order Processing System

Q1. Threading Issue: Deadlock.

Q2. Reason: circular wait. Thread A holds Inventory and waits for Payment while Thread B holds Payment and waits for Inventory, so neither can proceed (mutual exclusion, hold and wait, no preemption, circular wait all hold).

Q3. Solution: make every thread acquire the locks in the same fixed order (Inventory then Payment), or use try-lock with timeout, or deadlock detection and recovery.